Quantification of MoC Type C in Probability of Default Models

On the Aggregate Effect of Regulatory Requirements on Conservatism Levels

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Margin of Conservatism in Probability of Default Modeling

- After completing Probability of Default (PD) model development, practitioners typically address overall model uncertainty by defining a so-called Margin of Conservatism (MoC).
- The European Banking Authority (EBA) defines three categories corresponding to distinct sources of uncertainty in the estimated parameters:
  1. Type A: uncertainty related to data and methodological deficiencies;
  2. Type B: uncertainty related to changes in risk appetite and internal processes;
  3. Type C: general estimation error, i.e., statistical uncertainty arising from parameter estimation.
- One of the most commonly applied MoC Type C relates to uncertainty in estimating the long-run average default rate (central tendency).
- According to the EBA PD & LGD Guidelines, for the purpose of determining PD estimates in the calibration process, institutions may consider either the long-run average default rate at the grade (pool) level or at the calibration segment level. Accordingly, a MoC Type C is expected to be quantified.

ECB Interpretation of MoC Type C

- In line with the EBA's requirements, the ECB, in its Guide to Internal Models (2025), paragraph 327, section *Model-Related MoC*, clarifies its understanding of the quantification of MoC Type C, emphasizing that specific sources of uncertainty should reflect the dispersion of statistical estimators.
- Furthermore, the ECB distinguishes between models that output rating grades or pools and models with continuous outputs.
- As this presentation focuses on the most common approach in PD modeling, where the final model output is a rating scale, the remainder is dedicated to the ECB's interpretation in this context.
- For models with rating grade outputs, the ECB emphasizes two main sources of uncertainty:
  1. the average of one-year default rates over time;
  2. the number of exposures per rating grade and per year, which affects the uncertainty of the one-year default rate.

ECB Interpretation of MoC Type C cont.

- Additionally, with regard to the average of one-year default rates, the ECB notes that dependency between default rates over time should be considered in the MoC Type C quantification process.

Notes

- Given these clarifications from the ECB, and taking into account relatively short time series as well as specific statistical constraints in estimating the dependency structure, it is questionable how accurately all these sources can be incorporated into the quantification of MoC Type C.
- Moreover, it is critical to understand the contribution of each source of uncertainty to the overall level of MoC Type C in order to properly balance the selected methods and the underlying assumptions used for quantification.

Challenges in Quantifying MoC Type C

- The ECB has set clear expectations for MoC Type C quantification, stating that the fewer the observations per grade and the shorter the time series, the higher the Type C assigned to that grade should be. Additionally, the ECB expects that the correlation between yearly default rates is properly addressed in this process.
- In practice, institutions often use simplified frameworks for MoC Type C quantification, typically neglecting either the correlation structure of default rates over time, the variability within specific reference dates, or both. In some approaches, the variability of default rates over time is also not taken into account.
- Considering some or all sources of uncertainty, combined with different statistical approaches, can result in significantly different MoC Type C levels being added to the PD best estimate.
- The above raises several questions that practitioners face in this exercise:
    - Should every simplification of quantification methods and assumptions be explicitly considered as regulatory non-compliance?
    - Which sources of uncertainty should always be considered?
    - What is the appropriate level of MoC Type C given the relatively short time series used for its quantification?
- The purpose of this presentation is to compare some of the statistical methods used in practice for the quantification of MoC Type C related to uncertainty in estimating the LRA. As the presented approaches are not exhaustive, practitioners are encouraged to consider other methods, but are advised to keep in mind the underlying assumptions and their alignment with regulatory expectations. The following slides outline the basics of each statistical approach, while the remainder of the presentation focuses on the simulation study and presents the results.

Statistical Method 1

This method assumes that the LRA default rate follows a normal distribution and accordingly calculates the upper bound of the selected confidence interval. Formally, this can be written as follows:

$$\text{LRA} + z_{1-\alpha} \frac{\sigma_{\text{ODR}}}{\sqrt{\text{n}}}$$

where:

- LRA is the long-run average calculated as the arithmetic average of reference dates' default rates;
- $z_{1-\alpha}$ is the $1-\alpha$ quantile of the standard normal distribution, where α is the significance level and $1-\alpha$ the selected confidence level;
- σ_{ODR} is the standard deviation of the observed reference dates' default rates;
- n is the number of observations (reference dates) used for the calculation of the LRA.

Note:

This method does not consider:

1. the uncertainty of the observed default rates within the reference dates;
2. the dependency of default rates over time.

Statistical Method 2

This method extends Method 1 by considering the effective sample size to account for the dependency of default rates over time. The following formula presents this adjustment:

$$\text{LRA} + z_{1-\alpha} \frac{\sigma_{\text{ODR}}}{\sqrt{\text{n}_e}}$$

where:

- LRA is the long-run average calculated as the arithmetic average of reference dates' default rates;
- $z_{1-\alpha}$ is the $1-\alpha$ quantile of the standard normal distribution, where α is the significance level and $1-\alpha$ the selected confidence level;
- σ_{ODR} is the standard deviation of the observed reference dates' default rates;
- n_e is the effective sample size obtained by dividing the actual length of the observed default rate series used for LRA calculation by $1 + 2\sum_{k=1}^{n} \text{AR}^k$, where AR is the first-order autocorrelation.

Note:

This method does not consider the uncertainty of the observed default rates within the reference dates.

Practitioners should note that the above formula assumes a first-order autocorrelation structure; however, the same approach can be extended to any assumed autocorrelation structure.

Statistical Method 3

This method is based on Jeffreys' confidence interval for a binomial proportion. For the selected confidence level, Jeffreys' interval is given as:

$$\beta_{1-\alpha}(\text{shape1} = \text{ND} + 0.5, \text{shape2} = \text{N} - \text{ND} + 0.5)$$

where:

- $\beta_{1-\alpha}$ is the $1-\alpha$ quantile of the beta distribution with parameters shape1 and shape2;
- α is the significance level and $1-\alpha$ the selected confidence level;
- ND is the number of defaults at the reference date;
- N is the number of exposures at the reference date.

Note:

For the selected confidence level, this method calculates the upper bound of the binomial proportion of the observed default rates and then recalculates the LRA. The difference between the observed and simulated LRA represents the MoC level.

This method does not consider:

1. the uncertainty in the observed default rates over time;
2. the dependency among default rates over time.

Statistical Method 4

This method is similar to Method 3 but assumes a normal approximation of the binomial proportion distribution and accordingly derives the upper bound of the confidence interval. Formally, this can be written as follows:

$$\text{ODR} + z_{1-\alpha}\sqrt{\frac{\text{ODR}\cdot(1-\text{ODR})}{N}}$$

where:

- ODR is the observed default rate at the reference date;
- $z_{1-\alpha}$ is the $1-\alpha$ quantile of the standard normal distribution, where α is the significance level and $1-\alpha$ the selected confidence level;
- N is the number of exposures at the reference date.

Note:

For the selected confidence level, this method calculates the upper bound of the binomial proportion of the observed default rates and then recalculates the LRA. The difference between the observed and simulated LRA represents the MoC level.

This method does not consider:

1. the uncertainty in the observed default rates over time;
2. the dependency among default rates over time.

Statistical Method 5 and 6

These two methods assume that the long-run average default rate follow a Vasicek distribution. The parameters of the Vasicek distribution - asset correlation (ρ) and average PD - can either be calculated from the observed default rates or, at a minimum, the asset correlation can be taken as the one used for capital requirement calculations.

For Method 5, the asset correlation is calculated from the observed data using a maximum likelihood approach, while for Method 6, it is assumed to be 0.10. In both methods, dependency over time is incorporated into the development of the systemic factor (z) by using first-order autocorrelation coefficients obtained from the observed data.

Finally, both approaches account for within-year uncertainty by assuming that the number of defaults follows a binomial distribution with a probability of default equal to the simulated PD from the Vasicek model, given the realization of the systemic factor. The formula below presents the main mechanism for simulating the systemic factor:

$$z_{t,t\leq T} = \phi z_{t-1} + \sqrt{1-\phi^2}\epsilon_t$$

where both $z_{t,t=1}$ and ϵ_t are drawn from the standard normal distribution.

Note:

These two methods incorporate all sources of uncertainty identified by the ECB, differing only in their asset correlation assumptions. Comparing the results of these methods can help assess the sensitivity of a specific model design to the values of asset correlation.

Detailed explanations of the underlying approach for both methods can be found here.

Practitioners are encouraged to consider other simulation-based methods, compare the results, and assess the level of MoC Type C under different underlying assumptions.

Simulation Study

The purpose of this simulation is to present results on the relative MoC Type C level depending on the selected statistical approach. Moreover, as each statistical approach relies on specific assumptions and/or considers different sources of uncertainty incorporated into the calculation, another aim of this simulation is to provide an overview of changes in MoC Type C levels. Specifically, it illustrates how the aggregation of particular sources of uncertainty, together with the selection of the statistical method, impacts the MoC Type C level.

For the sake of result comparability, the results are presented relative to the observed LRA (central tendency), which for the simulation data is recorded at 4.73%. Additionally, for methods that incorporate the dependency of default rates over time and asset correlation parameters, these are estimated at levels of 0.295 and 0.014, respectively. For the methods that consider uncertainty within the reference data, it is assumed that the number of exposures per year is equal and ranges from 100 to 100,000 (`N`).

Practitioners should note that the first-order autocorrelation coefficient is estimated based on the systemic factor from the Vasicek model and is therefore used across all applicable methods. This is done purely for simplicity and demonstration purposes; however, for certain methods, alternative designs may be more appropriate.

For full replicability of the results, the simulation dataset is available at this link.

As the presented statistical methods are not exhaustive, practitioners are encouraged to consider other possible approaches and compare the results.

Simulation Results

```
##  Method       N MoC_Relative_Increase   LRA
##       1     100                 9.28% 4.73%
##       2     100                12.57% 4.73%
##       3     100                94.39% 4.73%
##       4     100                73.18% 4.73%
##       5     100                21.50% 4.73%
##       6     100                39.46% 4.73%
##       1   1,000                 9.28% 4.73%
##       2   1,000                12.57% 4.73%
##       3   1,000                25.22% 4.73%
##       4   1,000                23.14% 4.73%
##       5   1,000                13.89% 4.73%
##       6   1,000                36.72% 4.73%
##       1  10,000                 9.28% 4.73%
##       2  10,000                12.57% 4.73%
##       3  10,000                 7.52% 4.73%
##       4  10,000                 7.32% 4.73%
##       5  10,000                12.79% 4.73%
##       6  10,000                36.50% 4.73%
##       1 100,000                 9.28% 4.73%
##       2 100,000                12.57% 4.73%
##       3 100,000                 2.33% 4.73%
##       4 100,000                 2.31% 4.73%
##       5 100,000                12.80% 4.73%
##       6 100,000                36.37% 4.73%
```

Discussion Points

- Given the significant differences in MoC Type C levels depending on the statistical approach and underlying assumptions, is there a single "best" method to properly address the quantification process?
- Can practitioners set a "reasonable" level of MoC Type C for a modeled portfolio type?
- How can practitioners balance regulatory requirements with a reasonable level of MoC Type C?
- Is it always reasonable to add MoC Type C to the best estimate?
- Can MoC Type C quantification rely entirely on a statistical approach?